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Effect of transcutaneous electrical acupoint stimulation for preemptive analgesia on intraoperative opioid dosage in burn patients: a randomized, double-blind, controlled trial

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Abstract

Background Burn survivors may be at risk of various opioid-related adverse effects. We aimed to investigate whether transcutaneous electrical acupoint stimulation (TEAS) used for preemptive analgesia in burn patients reduces intraoperative opioid dosage.

Methods In this trial, 86 burn patients with wound debridement under general anesthesia were randomized to receive thirty minutes of TEAS (TEAS group, $n = 43$) at acupoints LI4 (Hegu), PC6 (Neiguan), ST36 (Zusanli), and SP6 (Sanyinjiao) before induction of anesthesia or false stimulation (control group, $n = 43$). The primary outcome was the intraoperative remifentanil dosage. Secondary outcomes included intraoperative propofol dosage, patients' resting facial-visual analogue scale (F-VAS) scores before stimulation (F1), after stimulation (F2), at 24 h after surgery (F3), and 48 h after surgery (F4), the incidence of opioid-related adverse reactions, and the use of rescue analgesics until the first postoperative day (POD1).

Results The dosage of remifentanil showed a significant difference between the two groups, with 0.65(0.45,1.00) in the TEAS group (T group) and 0.90(0.67,1.23) in the control group (C group) ($P < 0.05$). Meanwhile, there was no significant difference in the intraoperative propofol dosage between groups. F2 and F3 in the T group were lower than those in the C group ($P < 0.05$), and no significant differences were observed in F1 and F4 ($P > 0.05$) between the two groups. Until POD1, the incidence of nausea-vomiting and constipation ($P < 0.05$) in the T group was lower than that the C group, while there was no significant difference in the incidence of drowsiness and dizziness ($P > 0.05$). It is worth noting that the use of rescue analgesics between the two groups was comparable ($P > 0.05$).

Conclusions The preoperative application of TEAS for preemptive analgesia can reduce intraoperative remifentanil dosage. Meanwhile, it has a positive effect on early postoperative analgesia and gastrointestinal adverse reactions.

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Trial registration This study was registered in the Chinese Clinical Trials Registry (ChiCTR2300078615, date of registration: 14/12/2023, retrospectively registered).

Keywords Transcutaneous electrical acupoint stimulation, Burns, Remifentanil, Preemptive analgesia

Introduction

Worldwide, burns have become a major public health problem affecting the health-care system (Boldeanu et al., 2020), and approximately 11 million people require medical intervention for burn injuries each year (James et al., 2020). Pain management in burn patients is challenging, as the pain from a burn injury can be excruciating, which is considered the most intense pain. Opioids are the primary medication for burn analgesia (Romanowski et al., 2020). However, the dosage and duration of opioids in burn patients are significantly higher than the guidelines' standards (Emery et al., 2017; Emery and Eitan 2020). Also, patients who have survived a burn injury might be at risk of opioid dependence (Mauck et al., 2024; Jamal et al., 2024). Despite the strong analgesic effect of opioids, it is far from benign, including numerous adverse effects (Paul et al., 2021). Most burn patients require multiple wound debridements, undergoing general anesthesia, thus providing comfortable analgesia for them while reducing the dosages of opioid will be a major challenge for anesthesiologists.

It is supported that preemptive analgesia, which means reducing peripheral and central nerve sensitization caused by afferent injurious stimulus before they act on the organism, can effectively relieve perioperative pain and reduce analgesic drug dosage (Taumberger et al., 2022; Xuan et al., 2022) and is more effective than analgesic treatment after the surgery (Isuru et al., 2017). As burn patients' pain has been triggered by local tissue damage and its induced inflammatory response after injury, followed by other pain associated with burn wound treatment. These persistent pain impulses are transmitted to the central nervous system, resulting in the lower patients' pain threshold and greater sensitivity to external stimulus (Tan et al., 2011). Therefore, burn patients have different degrees of pain and nociceptive sensitization before receiving debridement.

TEAS, an important non-invasive, safe medical aid measure widely used for perioperative analgesia (Szmit et al., 2023; Shah et al., 2022), was proven to alleviate pain in burn patients (Abali et al., 2015; Cuignet et al., 2015). From the periphery to the central nervous system, multiple molecular pathways, inflammatory factors, neurotransmitters are involved in its analgesic mechanism, including activation of the endogenous opioid peptide system (Fan et al., 2023), modulation of the expression of and activity of ion channels (Li et al., 2019), regulation of the balance of peripheral proinflammatory and anti-inflammatory cytokines (Wang et al., 2019), inhibition of activation of spinal cord neuroglial cells (Wang et al., 2018), modulation of pain-related signaling pathways involved in pain sensitization formation to alleviate nociceptive sensitization, and regulation of pain-related brain regions and cerebral circuits to exert analgesic effects (Ma et al., 2020). Thus, TEAS not only has analgesic effects but also has the effect of decreasing the peripheral and central sensitization (Lai et al., 2019), which is consistent with the concept of preemptive analgesia. More than that, compared with other analgesic measures, like regional and spinal anesthesia, and drug-based analgesia, TEAS has the advantage of being non-invasive, easy to operate, and having fewer adverse effects, including drug-related reactions, local anesthetic toxicity, and vascular and neurologic injury. At the same time, compared with other analgesic measures, TEAS has unique advantages in regulating patients' immune function, alleviating postoperative cognitive function and gastrointestinal adverse reactions (Li et al., 2021; Xi, et al., 2021; Cheng, et al., 2023), which is more in line with the concept of enhanced recovery after surgery (ERAS). However, TEAS used for preemptive analgesia to reduce intraoperative opioid dosage in burn patients has not been studied before, to our knowledge.

Therefore, this study aimed to investigate whether preoperative application of TEAS used for preemptive analgesia in burn patients reduces intraoperative opioid dosage and its impact on patients' postoperative recovery.

Methods

Patients and study design

This study protocol was approved by the Science and Technology Ethics Committee of the First Affiliated Hospital of Shihezi University on December 1, 2023 (reference: KJ2023-385–01) and registered in the Chinese Clinical Trials Registry (ChiCTR2300078615, date of registration: 14/12/2023, retrospectively registered). Between December 2023 and December 2024, American Society of Anesthesiologists (ASA) physical status I or II in moderate burn adult patients aged ≤ 60 years who were planned to undergo burn wound debridement under general anesthesia were enrolled in this study after providing informed consent. Patients were excluded if they had contraindications to TEAS, collateral risks attributable to pregnancy or breastfeeding, alcohol or substance abuse, chronic pain or long-term analgesic medication, or major systemic diseases.

Randomization and blinding

Patients were randomly assigned to either the TEAS group (T group) or the control group (C group), using a computer-generated design (with the time of the start of the study, 20,231,212 as the random number seed), in a 1:1 ratio. Allocations were concealed in sequentially numbered, opaque, sealed envelopes prepared by independent study monitors. After entering the preparatory room, the TEAS operator, who was not involved in the care of the patient and independent of this study, opened the numbered envelope, followed the arrangement, and conducted TEAS or false stimulation, but did not know the significance of the numbers. All other individuals involved in the study (patients, surgeons, anesthesiologists, data collectors, data analysts) were blinded to the randomization and the stimulation.

Intervention

After informing the patients of the steps of electrical stimulation, both groups pasted electrode sheets at the selected acupoints, and T group was stimulated by Han's acupoint nerve stimulator (HANS-200A, Nanjing Jisheng Medical Technology CO., LTD.) for 30 min before the induction of anesthesia, while C group was stimulated by the placebo-type Han's acupoint stimulator (Sham TEAS), which had no therapeutic effect. The two interventions both caused numbness in acupoints.

Acupoint selection

Bilateral LI4 (Hegu), PC6 (Neiguan), ST36 (Zusanli) and PC6 (Sanyinjiao).

Operating method

In the supine position, the skin surface of the acupoint was degreased with 75% alcohol and dried, and the gel electrode sheet was attached to the prescribed acupoints with tight pressing.

Parameters

Time, 30 min before anesthesia induction; Frequency, 2/100 Hz; Intensity, < 10 mA (gradually adjusted to the maximum intensity acceptable to the patient).

The patients were monitored to assess reactions to the stimulation. The adverse events (AE) include pallor, skin pigmentation, itching, vertigo, and chest tightness. The serious adverse events (SAE) refer to any medical events that may pose a threat to life, result in persistent disability or functional impairment, such as syncope, spasm, and cardiac emergencies (Hwan et al., 2020). In the event of a SAE, immediately cease the procedure and initiate emergency treatment to ensure the patient's safety. Report the SAE to the ethics committee and conduct follow-up until the completion of treatment.

Pain assessment

Facial-visual analogue scale (F-VAS), range of 0–100 mm, with one end representing "no pain at all (0 mm)" and the other end representing "extreme pain (100 mm)", was used to evaluate patients' perioperative resting pain. A higher score indicates more severe pain.

Anesthesia and analgesia

All patients routinely underwent cardiac monitoring, including blood pressure (BP), heart rate (HR), pulse oxygen saturation, and electrocardiogram. Both groups of patients were given total intravenous anesthesia, induction drugs: midazolam 0.05 mg/kg, sufentanil 0.5 μg/kg, etomidate 0.2 mg/kg, and rocuronium 0.7 mg/kg. Anesthesia maintenance drugs: propofol 4–12 mg/kg-h, remifentanil 0.1–1 μg/kg-min intravenous pumping, and rocuronium 0.2 mg/kg intravenous injection as needed; The intraoperative dynamic adjustment of propofol and remifentanil dosage was according to hemodynamics, bispectral Index (BIS) and surgical pleth index (SPI), so that BP and HR remained stable, BIS value should be maintained between 40–60 and SPI value between 20–50. Thirty minutes before the end of surgery, intravenous sufentanil 0.1 μg/kg was pushed, and the infusion of rocuronium bromide was stopped. The infusion of propofol and remifentanil was discontinued at the end of the operation.

All patients were treated with patient-controlled intravenous analgesia (PCIA) after surgery. The PCIA configuration method was as follows: sufentanil 100 μg + butorphanol 4 mg + dexamethasone 5 mg + ondansetron 8 mg + 0.9% sodium chloride, configured to a total of 100 ml. All patients were instructed by the same nurse to use the PCIA pump, and pressed the PCIA pump when F-VAS was above 40 mm, with an interval of 15 min; when pain was not effectively relieved by pressing the PCIA pump for more than 3 times, or F-VAS was above 50 mm, burn patients were given dezocine 5 mg intramuscularly as a remedial analgesic. If necessary, administer every 6 h, not exceeding 20 mg per day.

Observational indicators

General observational indicators: age; sex; body mass index (BMI); weight; height; burn areas; ASA grade; rates of hypertension, diabetes mellitus, coronary heart disease and obesity; mean arterial pressure (MAP) and HR 5 min before the TEAS intervention (T1), immediately after endotracheal intubation (T2), immediately after peeling (T3), 5 min before the end of surgery (T4) and immediately after endotracheal extubation (T5); operation time; anesthesia time.

Primary outcome indicators: intraoperative remifentanil dosage.

Secondary outcome indicators: intraoperative propofol dosage; patients' resting F-VAS scores before stimulation (F1), after stimulation (F2), at 24 h after surgery (F3), and 48 h after surgery (F4); the occurrence of nausea-vomiting, constipation, drowsiness, and dizziness, and the use of rescue analgesics until the first postoperative day (POD1).

Sample-size calculation

Based on pre-experimental results, the average intraoperative remifentanil dosage was estimated to be 0.15 μg/kg/min, and the standard deviation was 0.05. Based on the previous study (Wang et al., 2023), TEAS under general anesthesia may reduce intraoperative remifentanil consumption by approximately 20%. Thus, we considered the average medication dosage and standard deviation decreasing to 0.12 μg/kg/min and 0.03 as clinically relevant, with a power of 90% and significance of 5%, 82 participants would be required to detect the reduction. To compensate for dropouts, we increased the sample size of randomized patients to 102.

Statistical analyses

Statistical analyses were performed using SPSS 25.0 (IBM SPSS Statistics for Windows). Continuous variables were tested for normality Shapiro–Wilk test. The measurement data conforming to normal distribution were expressed as mean ± standard deviation ($\overline{x}$±s), and compared using independent samples t-tests between groups. Non-normally distributed data were expressed as median (M) and interquartile range (IQR) and compared using the Mann–Whitney U test between groups. Count data were expressed as numbers (n) with percentages (%) and analyzed by Pearson's χ^2 test or Fisher's exact test as appropriate. Comparison between groups of hemodynamic indicators (MAP, HR) was performed by repeated measure ANOVA. A *P*-value < 0.05 was considered statistically significant.

Results

Patient characteristics

Among the 102 patients, 16 were excluded: 9 patients declined to participate, 5 patients withdrew from the study due to surgical modification, and one patient in each of the two groups withdrew due to redness and itching on the skin while applying electrodes. Finally, 86 patients were enrolled in this study, with 43 patients in each group (Fig. 1). There were no SAEs in the study groups.

The baseline patients' characteristics were comparable between the two groups (P > 0.05), and there was no difference in surgical data between the groups (P > 0.05). (Table 1, Figs. 2, 3).

Primary outcome

The dosage of remifentanil showed a significantly difference between the two groups (P = 0.039), with 0.70(0.50,1.10) in T group and 0.90 (0.65,1.25) in C group (Fig. 4).

Secondary outcomes

The consumption of intraoperative propofol was not significantly different between the two groups (P = 0.3) (Table 2).

Before the intervention, F1 in T group and C group was comparable (P = 0.336). After the intervention, there was a significant decrease in F-VAS scores in both groups (P < 0.05). Meanwhile, compared with C group, F2 was significantly reduced in T group (P = 0.030). At 24 h after surgery, F3 was found to be significantly lower in T group compared to C group (P = 0.039), but at 48 h after surgery, F4 was similar between the two groups (P = 0.061) (Fig. 5).

The incidence of nausea-vomiting (P = 0.011) and constipation (P = 0.033) until POD1 in T group was significantly lower than that in C group, but there was no significant difference observed in the incidence of drowsiness (P = 0.178) and dizziness (P = 0.176) (Table 2). All adverse reactions resolved spontaneously or with symptomatic treatment following evaluation.

Until POD1, the use of rescue analgesics was similar between the two groups (P = 0.430) (Table 2). All patients receiving rescue analgesia had pain scores reduced to below 50 mm after receiving 5 mg of dezocine for intramuscular injection, and no second rescue analgesia was administered.

Discussion

This study indicates that TEAS used for preemptive analgesia significantly reduces intraoperative opioid consumption in burn patients, alleviates early postoperative pain, mitigates postoperative gastrointestinal adverse reactions, and promotes rapid recovery.

As an important medical auxiliary measure, TEAS has been widely used in alleviating perioperative pain (Tan et al., 2024). However, acupoint parameters vary across different studies. According to the median theory of traditional Chinese medicine, LI4, PC6, and ST36 are the main acupoints for analgesia (Han 2011). The analgesic effect of LI4 is widely used in various clinical departments to treat pain-related diseases. PC6 is often used to treat a variety of pain symptoms and has the effect of tranquilizing the mind. ST36 is mainly used for treating stomach pain, abdominal pain, headache, and paralyzing pain in the lower limbs. Appropriate stimulation of SP6 can not only effectively alleviate the pain of the lower limbs, but also be effective for the treatment of gastrointestinal diseases.

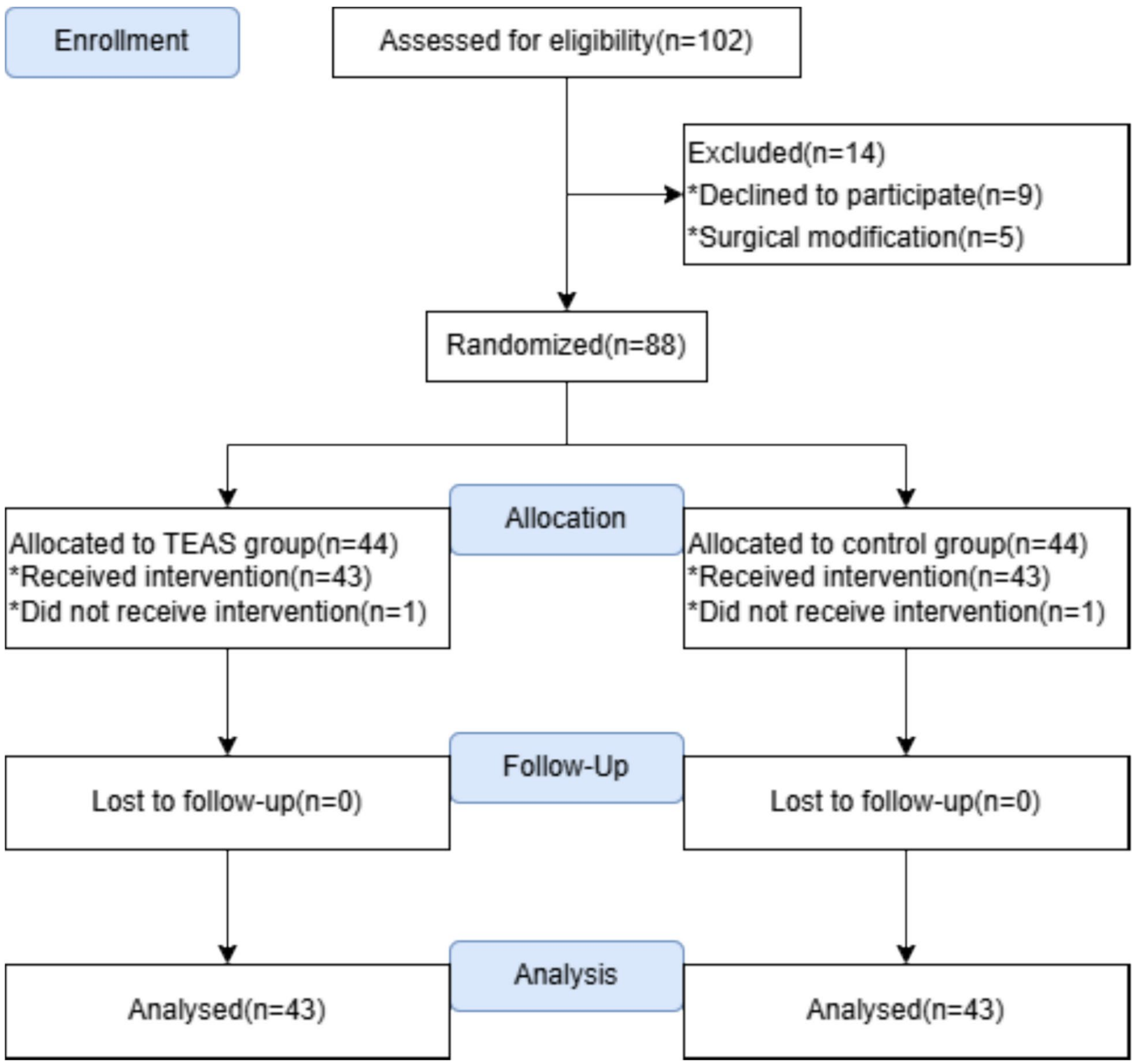

Fig. 1 CONSORT flow diagram illustrating the study design and the inclusion/exclusion patient population flow. CONSORT Consolidated Standards of Reporting Trials. TEAS, transcutaneous electrical acupoint stimulation

The results of this study found that the intraoperative remifentanil dosage in T group was lower than that in C group, while the dosage of propofol in the study groups was similar, echoing similar results found in other previous research (Wang et al., 2023). Although resting F-VAS scores in both groups decreased after the intervention, F2 in T group was significantly lower compared with C group. The decrease in F-VAS scores in C group may be associated with a placebo effect. It indicates that TEAS reduces the intraoperative dosage of opioids, which may be related to its role in preemptive analgesia. TEAS can cause the nervous system to release endogenous opioid peptides, to reduce the sensitization of central neurons to provide pain relief (Fan et al., 2023; Qiao et al., 2020). Apart from this, burn injury profoundly alters the functional state of the immune system, which is the reason explaining the reduced potency of opioids (Emery et al., 2017; Emery and Eitan 2020). Therefore, the effect of decreasing intraoperative opioid dosage may also be associated with controlling the systemic inflammatory response (Que et al., 2021) and reducing opioid resistance to injury in burn patients. Even though TEAS has been confirmed to have a sedative effect (Zhang et al., 2024; Lu et al., 2022), the use of intraoperative sedatives was similar between the two groups. This may be related to the fact that acupoints strongly associated with sedation were not selected for this study, or that the sedative effect of TEAS didn't carry over into the operation.

Table 1 Comparison of baseline characteristics between the two groups

Characteristics	T group($n=43$)	C group($n=43$)	*P*-value
Age (year)	42.0 ± 9.8	39.5 ± 11.3	0.277
BMI (kg/m^2)	24.6 ± 3.5	24.9 ± 3.2	0.720
Weight (kg)	71.9 ± 13.4	71.8 ± 12.6	0.980
Height (cm)	170.5 ± 8.0	169.5 ± 7.4	0.602
Burn areas (%)	14.0(13.0,17.0)	14.0(13.0,16.0)	0.626
ASA grading, *n* (%)			0.610
I	11(25.6)	9(20.9)	
II	32(74.4)	34(79.1)	
Sex, *n* (%)			0.795
Female	9(20.9)	10(23.3)	
Male	34(79.1)	33(76.7)	
Hypertension, *n* (%)			0.802
Yes	10(23.3)	11(25.6)	
No	33(76.7)	32(74.4)	
Diabetes Mellitus, *n* (%)			0.763
Yes	6(14.0)	7(16.3)	
No	37(86.0)	36(83.7)	
Coronary Heart Disease, *n* (%)			> 0.999
Yes	4(9.3)	5(11.6)	
No	39(90.7)	38(88.4)	
Obesity, *n* (%)			> 0.999
Yes	6(14.0)	6(14.0)	
No	37(86.0)	37(86.0)	
Operation time (min)	88.4 ± 35.6	89.0 ± 33.6	0.938
Anesthesia time (min)	100.9 ± 36.3	101.4 ± 33.9	0.951

Characteristic presented as mean ± SD or M and IQR or (n, %) of patients

SD Standard-deviation, *M* Median, *IQR* Interquartile range, *BMI* Body mass index, *ASA* American Society of Anesthesiologists

At 24 h postoperatively, F3 in T group was found to be lower than that in C group, but there was no significant difference in F4 at 48 h after surgery between the two groups. It suggests that TEAS used for preemptive analgesia may have a more noticeable impact on pain relief during the early postoperative phase, which is consistent with previous studies (Tu et al., 2019). Since preemptive analgesia typically involves pain intervention before surgical incision to block central sensitization, but pain-related neural sensitization is a continuous process throughout the perioperative period, wound injury may re-initiate central sensitization once the blockade of nociceptive afferent stimulation is lifted. This may explain why pain scores remained comparable between the two groups at 48 h after surgery. Preventive analgesia emphasizes sustained suppression of nociceptive signal transmission throughout the perioperative period to minimize central sensitization (Clarke et al., 2015). Future studies may explore preventive analgesia targeting TEAS to assess its impact on patients' long-term quality of life and pain scores.

At the same time, the use of rescue analgesics in both groups was low, and no significant difference was observed between the study groups, which contradicts previous research findings (Yan et al., 2025; Liu et al., 2024). On the one hand, we suspect that pain is not intense enough to require remedial analgesia after the use of PCIA. On the other hand, we controlled the value of SPI between 20 and 50, which may be correlated to the decrease in the patients' postoperative pain level and the decrease in the need for analgesic drugs (Hung et al., 2022; Jung et al., 2020).

Opioids have been reported to have quite a few adverse effects. In our study, we found that the incidence of nausea-vomiting and constipation until POD1

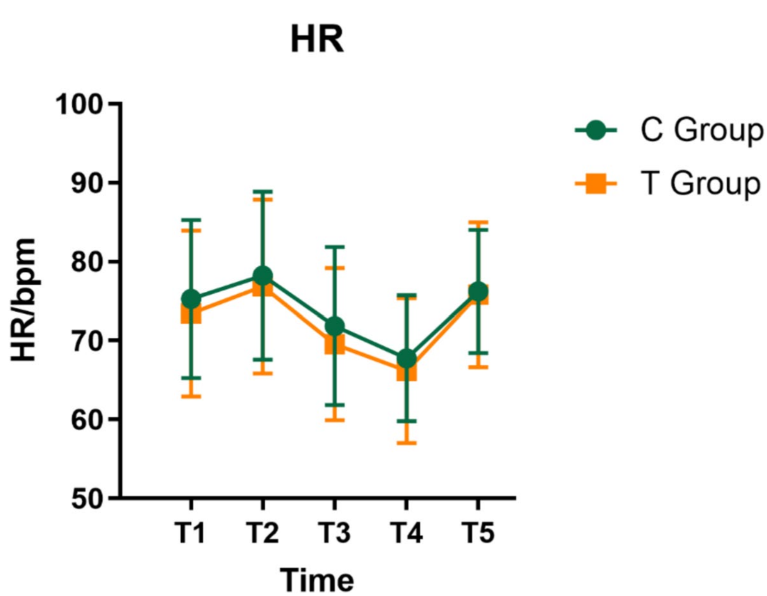

Fig. 2 Comparison of HR between the two groups at distinct junctures. Note: Data are presented as the mean ± SD

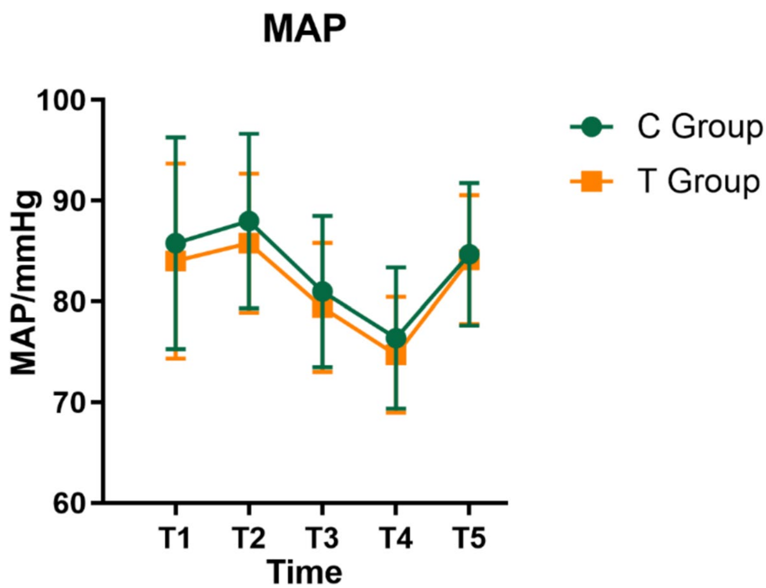

Fig. 3 Comparison of MAP between the two groups at distinct junctures. Note: Data are presented as the mean ± SD

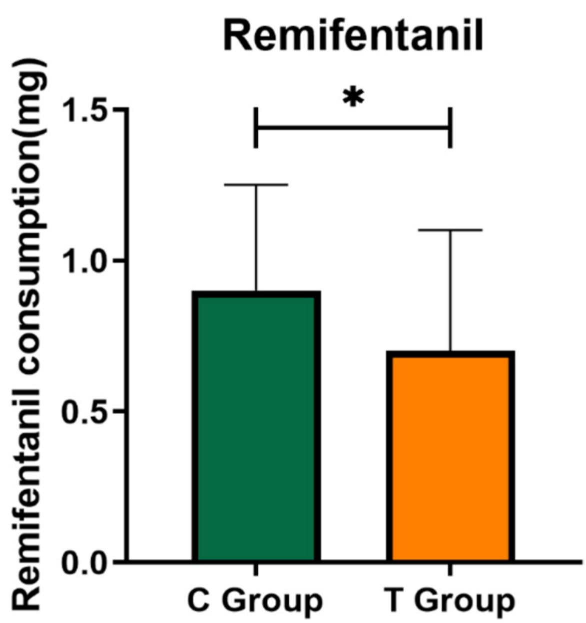

Fig. 4 Comparison of remifentanil consumption between the two groups. Note: Data are presented as the median with interquartile range. * indicates a significant decrease in intraoperative remifentanil consumption in group T compared to group C ($P < 0.05$)

Table 2 Comparison of secondary outcomes between the two groups

Secondary outcomes	T group($n = 43$)	C group($n = 43$)	*P*-value
Propofol consumption (mg)	400.0(305.0,625.0)	450.0(365.0,720.0)	0.296
Nausea-vomiting, n (%)	7(16.3)	16(37.2)	0.028
Constipation, n (%)	8(18.6)	17(39.5)	0.033
Drowsiness, n (%)	3(7.0)	7(16.3)	0.178
Dizziness, n (%)	6(14.0)	11(25.6)	0.176
Use of rescue analgesics, n (%)	3(7.0)	8(18.6)	0.106

Secondary outcomes presented as M and IQR or n(%) of patients. M-median

was significantly decreased in T group compared with C group. However, there was no significant difference in drowsiness and dizziness. The situation is consistent with previous studies (Zhou et al., 2024; Mao et al., 2021). Considering this situation, we guess that the reduction of nausea-vomiting and constipation by TEAS was not accomplished by decreasing the intraoperative opioid dosage, but rather by its other potential mechanisms, including modulating neurotransmitters, affecting the central nervous system, reducing vagal action, affecting chemoreceptors, and reducing inflammatory response (Li et al., 2024).

Most of the previous studies adjusted intraoperative opioid dosage according to blood pressure and heart rate changes or the anesthesiologists' experience, which has greater limitations and subjectivity. In this study, we used the BIS value, a monitoring index of the depth of sedation, and SPI, an analgesic index that has been widely used for the evaluation of surgical patients (Oh et al., 2024), to guide intraoperative sedation and analgesia administration. This approach enhances the reliability of the research.

This study also has certain limitations that should be acknowledged. As this was a single-center study, the sample size was small, and a multi-center study is needed to validate the findings. Moreover, there are no objective laboratory indicators available to elaborate on outcome indicators and composite recovery profiles employed to

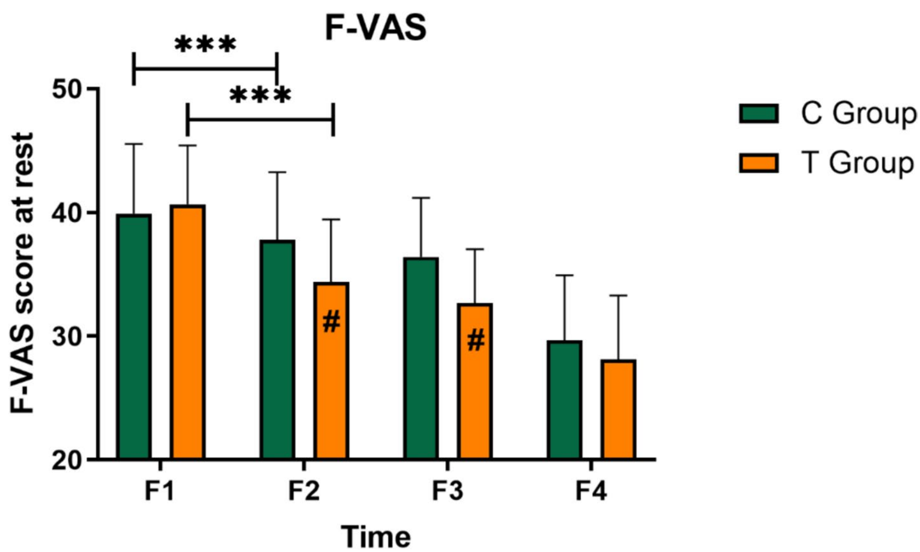

Fig. 5 Comparison of resting F-VAS scores between the two groups at the same point. Note: Data were presented as mean ± SD. # indicates that the F-VAS scores were significantly lower in group T than in group C ($P < 0.05$). *** indicates a significant decrease in F-VAS scores after intervention in both groups ($P < 0.001$)

evaluate patients' postoperative recovery. Ultimately, the follow-up of the patients in this study was only conducted 48 h after surgery, and the postoperative recovery of the patients was not followed up for a long-time, so the long-term effect of TEAS needs further research.

Strengths and limitations

The use of BIS (Bispectral Index) and SPI (Surgical Pleth Index) make our results more objective and reliable. This study also has certain limitations that should be acknowledged. As this was a single-center study, the sample size was small, and a multi-center study is needed to validate the findings. Moreover, there are no objective laboratory indicators available to elaborate on outcome indicators and composite recovery profiles employed to evaluate patients' postoperative recovery. Ultimately, the follow-up of the patients in this study was only conducted 48 h after surgery, and the postoperative recovery of the patients was not followed up for a long time, so the long term effect of TEAS needs further researched.

Conclusion

The preoperative application of TEAS for preemptive analgesia can reduce intraoperative remifentanil dosage. Meanwhile, it has a positive effect on early postoperative analgesia and gastrointestinal adverse reactions.

Abbreviations

TEAS	Transcutaneous Electrical Acupoint Stimulation
F-VAS	Facial-Visual Analogue Scale
PCIA	Patient-Controlled Intravenous Analgesia
ASA	American Society of Anesthesiologists
BMI	Body Mass Index
BP	Blood Pressure
MAP	Mean Arterial Pressure
HR	Heart Rate
BIS	Bispectral Index
SPI	Surgical Pleth Index
POD1	The First Postoperative Day
AE	Adverse Events
SAE	Serious Adverse Events

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Authors' contributions

Study design: Qian Liu, Jiangwen Yin, Yan Li. Study conduct: Qian Liu, Ziyan Pan, Haichuan He, Yujia Zhao. Data analysis: Qian Liu, Jingbin Zhang. Writing paper: Qian Liu, Ziyan Pan, Jingbin Zhang, Haichuan He, Jiangwen Yin, Yan Li. All authors read and approved the final manuscript.

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Data availability

The datasets used and analyzed during this current study are available from the corresponding author on reasonable request.

Declarations

Ethics approval and consent to participate

This randomized, double-blind, controlled trial was conducted in the First Affiliated Hospital of Shihezi University between December 2023 and December 2024. The study protocol was approved by the Science and Technology Ethics Committee of the First Affiliated Hospital of Shihezi University on December 1, 2023(reference: KJ2023-385–01) (Appendix 2) and registered in the Chinese Clinical Trials Registry (ChiCTR2300078615, date of registration: 14/12/2023). All participants provided written informed consent.

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

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